RoFormer: Enhanced Transformer with Rotary Position Embedding

Rotary Position Embedding

Absolute position embedding commonly adds p_i to x_i before projection:

$$f_t(x_i, i) := W_t (x_i + p_i), \quad t \in \{q, k, v\}.$$

RoFormer instead incorporates position by rotating query and key vectors.
For a two-dimensional subspace:

$$f_q(x_m, m) = R(m \theta) W_q x_m$$

$$f_k(x_n, n) = R(n \theta) W_k x_n$$

where $R(\phi)$ is [[cos phi, -sin phi], [sin phi, cos phi]].

The resulting dot product depends on relative phase $(m - n) \theta$.
For higher dimensions, independent 2D subspaces use frequencies θ_i.
Equivalently, adjacent real coordinates form complex components multiplied by $\exp(i m \theta_i)$.

During incremental decoding, m is the absolute token index.